

Venkata Varasiddi Manoj Kumar Jala

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EDUCATION

Indiana University, Bloomington

Master of Science in Computer Science, GPA:3.97/4

Bloomington, IN

Aug. 2023 – May 2025

SRM University

Bachelor of Technology, GPA:8.72/10

Chennai, India

Aug. 2017 – May 2021

EXPERIENCE

Software Developer Engineer

June 2021 – July 2023

Verizon

- Developed and deployed highly scalable Java microservices using Spring Boot and WebFlux, handling 500K+ concurrent requests daily and reducing API latency by 40% through reactive programming optimizations.
- Architected and implemented **RESTful APIs and GraphQL services** for finance, payments, and inventory systems, enabling real-time, efficient, and secure data exchange between microservices. **Designed a schema-based GraphQL solution** to optimize complex data queries, **reducing over-fetching and under-fetching issues by 30%**.
- **Developed reusable UI components using ReactJS, TailwindCSS, saga.js and Redux**, enhancing front-end performance and maintainability.
- Implemented **Redis caching** and optimized database indexing to **accelerate API response times by 35%**, reducing redundant database queries and enhancing system scalability.
- Implemented **event-driven microservices** communication using **Kafka and RabbitMQ**, processing 1M+ events/day with fault-tolerant message delivery.
- Utilized **ELK stack** (Elasticsearch, Logstash, Kibana) for real-time log monitoring and analytics, enabling proactive issue detection and **reducing mean time to detect and resolve critical issues by 50%**.
- **Increased test coverage to 90%** and reduced production defects by 60% by writing and executing automated tests using **JUnit, Jest, and JaCoCo**, ensuring robust functionality and stability across services.
- Led and contributed to **Agile methodology** (sprint planning, code reviews, backlog refinement), ensuring timely feature delivery and improving development velocity.
- Maintained high code quality using SonarQube, **fixing 1,500+ security vulnerabilities and ensuring 99% uptime** through proactive performance tuning.
- Designed and implemented **CI/CD pipelines with Jenkins**, streamlining the deployment process and reducing integration issues.

PROJECTS

AI Hiring Assistant Platform | *React, Tailwind CSS, FastAPI, CrewAI (multi-agent system), Vapi (voice API)*

- Developed a full-stack voice-enabled interview platform supporting **multi-role login** (Recruiter, Moderator, Interviewee) and personalized views.
- Designed recruiter dashboard to display **job-wise processing results**, moderator dashboard for **resume/interview tracking**, and a chatbot for candidates.
- Integrated **FastAPI** backend serving dynamic log data via polling.
- Implemented live logs and job status visualization with **auto-refresh and state-preserving UI updates**.

GitHub: github.com/manojkumarjala-dev/ai-hiring-assistant

BotBazaar – Robot Rental Platform | *MERN (MongoDB, Express.js, React, Node.js), Redux, OAuth, Google Authentication*

- Developed a full-stack web application enabling robot rentals with **role-based access** for clients, owners, and renters, along with admin oversight.
- Implemented authentication and security features, including **OAuth, Google Login, password reset, security questions, and two-factor protection**.
- Designed and managed product lifecycle workflows, supporting **CRUD operations**, session management with **Redux**, and secure transactions.

GitHub: github.com/manojkumarjala-dev/rental-management-system

FocusPal – Productivity & Habit Tracking App | *React Native, Expo, Firebase, Context API*

- Developed a cross-platform mobile app using **React Native** and **Expo**, enabling users to track habits, manage tasks, and enhance focus with a **Pomodoro timer**.
- Implemented **Firebase authentication** and **real-time database** to store and sync user data seamlessly across devices.
- Designed an intuitive UI with **React Native Paper** and **Styled Components**, improving user experience and engagement.

GitHub: github.com/manojkumarjala-dev/FocusPal

Sensor Network Web Application | *Next.js, NeonDB (PostgreSQL), Mapbox, Chart.js*

- Developed an interactive web application using **Next.js** and **NeonDB**, enabling real-time retrieval and processing of environmental sensor data from the **Healthy Cities Lab at IU**.
- Integrated **Mapbox** for geospatial visualization, displaying live sensor readings such as temperature, humidity, and air quality on an interactive map.
- Implemented dynamic data visualizations with **Chart.js**, allowing users to analyze environmental trends and compare sensor data across different locations.

GitHub: github.com/manojkumarjala-dev/sensor

Next.js Authentication Starter Pack | *Next.js, Auth.js, MongoDB*

- Developed a reusable authentication boilerplate using **Next.js**, **Auth.js**, and **MongoDB**, allowing seamless integration of user login, registration, and session management.
- Pre-configured **OAuth authentication** with providers like Google, enabling quick and secure authentication setup.
- Designed for scalability and ease of use, streamlining authentication implementation for new projects with minimal configuration.

GitHub: github.com/manojkumarjala-dev/nextjs-authjs-mongoDB

Distributed File Storage System | *Java, Spring Boot, AWS S3, Kafka, Elasticsearch, PostgreSQL, Spring Security*

- Developing a scalable **distributed file storage system** using **Spring Boot**, enabling secure file uploads, storage, and retrieval.
- Planning to implement **event-driven architecture** with **Kafka** for system notifications and logging.
- Integrating a **logging and monitoring service** using **Elasticsearch**, providing real-time tracking and audit capabilities.
- Developing a **notification service** for user updates, enhancing usability and system interactions.
- Implementing **Spring Security** for user authentication, ensuring **role based access control** and data protection.

GitHub: <https://github.com/manojkumarjala-dev/Distributed-File-Storage-System>

TECHNICAL SKILLS

Programming Languages: Java (8,11), Python, TypeScript, C++, JavaScript, HTML5/CSS3

Frameworks: Spring, Spring Boot, ReactJS, Redux, SQL, Node.js, Express, Flask, FastAPI, Selenium, Spring Security, Spring WebFlux

Developer Tools & Paradigms: Git, GitHub, Jenkins, AWS CodePipeline, Docker, Nginx, Tomcat, Maven, Gradle, npm, Postman, SonarQube, Jira, Elastic Stack, IntelliJ IDEA, VS Code, PyCharm, Eclipse, Unix, Reactive Programming, Object-Oriented Programming, Design Patterns, Data Structures

Databases: PostgreSQL, Oracle, MySQL, MongoDB, GraphQL, AWS DynamoDB, Cassandra, Cloudant, SQLite

Cloud Technologies: AWS (S3, EC2, Lambda, CloudFormation, CloudWatch, RDS, IAM, API Gateway, SQS, SNS), Google Cloud Platform (GCP), Azure, Kubernetes, Terraform